OlistDelayLens: A Deep Dive into Logistics Efficiency and Delivery Latency in Brazilian E-Commerce

Comprehensive Project Report

Sector: Logistics, Supply Chain, and E-Commerce Analytics

Team ID: Group E-G2

Name	Role	Primary Responsibility
Harsith	Project Lead	Repo setup, ETL, Timeline Mgmt
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Aradhya	Analysis Lead	EDA, Statistical Analysis, Quality
Ishan	Visualization Lead	Dashboard Design (Tableau)
Khushi	Strategy Lead	KPI Framework, Strategy, Recommendations
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1 EXECUTIVE SUMMARY

The "OlistDelayLens" project is an end-to-end data science and visual analytics initiative designed to untangle the complexities of e-commerce logistics within the Brazilian market. Utilizing a dataset of 100,000 orders from Olist (2016–2018), this report addresses the critical business challenge of delivery delays and their subsequent impact on customer trust and brand equity.

The Problem: Brazil's vast geography and fragmented logistics infrastructure lead to significant delivery variance. Olist, acting as a bridge between small sellers and large marketplaces, suffers when sellers fail to dispatch on time or when carriers underperform. Our analysis confirms that "Delivery Delay" is the single most significant predictor of negative customer sentiment, far outweighing product price or shipping cost.

Methodology: Our team developed a robust Python-based ETL pipeline that synchronized eight disparate datasets into a unified analytical master file. We performed rigorous data cleaning—handling 3,000+ missing records, removing extreme outliers (deliveries ¿53 days), and engineering spatial features using Haversine distance calculations. Statistical validation included T-Tests for satisfaction variance, ANOVA for regional performance bias, and Chi-Square tests for review score correlations.

Key Findings:

- **Growing Inefficiency:** The late delivery rate rose from 5.6% to 8.1% over 12 months, indicating that logistics operations are not scaling at the same pace as sales.
- **The Satisfaction Cliff:** On-time orders average a 4.2/5.0 review score, while late orders plummet to 2.4/5.0, representing a massive loss in potential customer lifetime value.
- **Regional Bottlenecks:** The state of Maranhão (MA) and Alagoas (AL) emerged as high-risk hubs with late rates exceeding 25% on specific routes.

Strategic Recommendations: We advocate for a multi-pronged approach: the enforcement of Seller SLAs with financial penalties for late dispatch, the implementation of dynamic "Buffer-Adjusted" ETAs based on geographic risk, and the decentralization of the logistics network through regional consolidation centers in high-risk zones.

2 SECTOR AND BUSINESS CONTEXT

2.1 Sector Overview: The Brazilian E-Commerce Landscape

The Brazilian e-commerce market is characterized by unique logistical challenges. Unlike North America or Europe, Brazil relies heavily on a mix of the national postal service (Correios) and private logistics firms to navigate a territory spanning 8.5 million square kilometers. Olist, a prominent marketplace integrator, operates as a platform that allows small and medium enterprises (SMEs) to sell on major retail sites. Because Olist does not own the inventory or the delivery fleet, they occupy a "vulnerable middle" where their reputation is at the mercy of both independent sellers and third-party carriers.

2.2 Stakeholders and Decision-Makers

The insights generated by this report are targeted at three distinct levels of leadership:

1. **Executive Leadership (COO/VP Logistics):** Focused on the "Executive Pulse"—macro trends in late delivery rates and the overall erosion of customer satisfaction.
2. **Regional Logistics Managers:** Interested in the "Geographic Risk Monitor" to identify specific carrier underperformance on critical routes (e.g., SP to NE routes).
3. **Seller Success Teams:** Focused on "Seller Accountability" to identify high-risk sellers who are consistently failing to meet dispatch windows.

2.3 Why This Problem Matters

In the modern e-commerce economy, "delivery speed" has shifted from a luxury to a baseline expectation. For a company like Olist, every delayed package is not just a logistical failure but a financial liability. Our data shows that a single day of delay can trigger a transition from a 5-star review to a 1-star review, which negatively impacts the search ranking of the seller and the overall trust in the Olist brand. By solving for delay, Olist can improve seller retention and increase customer repeat purchase rates.

3 PROBLEM STATEMENT AND OBJECTIVES

3.1 Formal Problem Definition

How can Olist leverage its historical transaction and logistics data to identify systemic bottlenecks, quantify the damage caused by delivery delays on customer satisfaction, and develop a data-driven strategy to improve delivery reliability across the Brazilian territory?

3.2 Project Scope

The scope of this project is confined to:

- **Dataset:** 99,441 unique orders from 2016 to 2018.
- **Geography:** All 27 Brazilian states (UF).
- **Logistics Cycle:** From order purchase to delivery at the customer's door.
- **Exclusions:** Financial payment methods, marketing spend analysis, and product quality issues unrelated to logistics.

3.3 Success Criteria

We define success through the following deliverables:

1. **Analytical Rigor:** Completion of a Python pipeline that cleans and transforms raw logs into an analysis-ready format.
2. **Actionable Insights:** Identification of at least five high-risk geographic routes and top 5% of underperforming sellers.
3. **Visual Utility:** Delivery of four interactive dashboards that allow stakeholders to drill down from national averages to individual seller performance.
4. **Statistical Evidence:** Proving the causal-link between delivery latency and customer satisfaction scores using hypothesis testing.

4 DATA DESCRIPTION

4.1 Source Citation

The primary data source is the "Brazilian E-Commerce Public Dataset by Olist" published on Kaggle. This dataset contains information on 100,000 orders made at multiple marketplaces in Brazil. **Access Link:** `https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce`

4.2 Dataset Structure and Coverage

The raw data is provided in a relational format across several files:

- **olist_orders_dataset**: The spine of the analysis, containing status and timestamps.
- **olist_order_items_dataset**: Item-level details, shipping limit dates, and freight values.
- **olist_customers_dataset**: Customer location (Zip code, City, State).
- **olist_sellers_dataset**: Seller location (Zip code, City, State).
- **olist_products_dataset**: Product weight and dimensions (critical for logistics volume).
- **olist_geolocation_dataset**: Lat/Lng coordinates for Haversine distance calculations.
- **olist_order_reviews_dataset**: The primary outcome variable (Review Score).

4.3 Key Features for Analysis

- **Timestamps**: Purchase, Approval, Carrier Handover, and Final Delivery.
- **Estimated Date**: The promise date given to the customer.
- **Geography**: Origin (Seller State) vs. Destination (Customer State).

4.4 Data Quality Challenges

The dataset presented several hurdles:

- **Missing Values**: 3.2% of orders were "canceled" or "shipped" but not delivered. These were removed to ensure we only analyzed completed logistics cycles.
- **Ambiguous Timestamps**: Some orders had approval dates earlier than purchase dates (system glitches).
- **Geolocation Complexity**: The raw geolocation data contained 1 million points. We aggregated this to the "Zip Code Prefix" level to prevent join-explosion.

5 CLEANING AND TRANSFORMATION (ETL)

5.1 The Python Extraction Pipeline

We implemented a multi-stage ETL process using the `Pandas` and `NumPy` libraries.

1. **Loading & Translation**: Raw Portuguese product categories were mapped to English for better report interpretability.
2. **Sequential Merging**: We performed a series of "Left Joins" starting from the Orders table, ensuring that item-level details and review scores were correctly aligned with order timestamps.
3. **Geographic Mapping**: A "Double Join" was performed on the Geolocation data—once for the Seller and once for the Customer—to attach coordinates for both ends of the shipment.

5.2 Major Cleaning Steps and Reasoning

- **Standardization**: All timestamp columns were cast to `datetime64[ns]` objects. This allowed us to perform time-delta calculations (e.g., `Delivery_Days = Delivered_Date - Purchase_Date`).
- **Median Imputation**: For missing product weight/dimensions, we used the median of the product category. This is more robust than a global mean as furniture is naturally heavier than electronics.
- **Outlier Trimming**: We identified that 0.5% of orders took over 53 days. These were usually lost packages or system errors and were excluded to prevent skewing our average performance metrics.

5.3 Feature Engineering for Logistics

- **Actual Delivery Time (Days)**: Total duration from purchase to customer receipt.
- **Delivery Delay**: `Actual_Time - Estimated_Time`. Positive values indicate a late arrival.
- **Haversine Distance**: We calculated the great-circle distance (km) between seller and customer. This is a crucial control variable; we cannot penalize a carrier for a 10-day delivery if the distance is 3,000 km, but we must investigate if the distance is only 20 km.

6 KPI FRAMEWORK

To drive business decisions, we established five core metrics:

1. **Late Delivery Rate (%)**
   - *Definition:* Percentage of delivered orders where the actual delivery date surpassed the estimated delivery date.
   - *Importance:* This is the "North Star" metric for Olist's brand reputation.
2. **Average Delay Magnitude (Days)**
   - *Definition:* The mean number of days by which late orders exceeded their ETA.
   - *Importance:* Tells us if we are failing by "hours" or "weeks".
3. **Dispatch Latency (Days)**
   - *Definition:* Time from order approval to carrier handover.
   - *Importance:* Isolates the seller's performance from the carrier's performance.
4. **NPS Loss (Satisfaction Gap)**
   - *Definition:* The difference in average review scores between "On-Time" and "Late" orders.
   - *Importance:* Quantifies the "Customer Experience Cost" of a logistical failure.
5. **Route Risk Index**
   - *Definition:* A composite score of distance vs. delay variance for specific state-to-state pairings.
   - *Importance:* Highlights structural bottlenecks in the national transit network.

7 EXPLORATORY ANALYSIS (EDA)

7.1 Temporal Trends

Our analysis revealed a worrying trend: while Olist's volume grew by over 100% between 2017 and 2018, the average late rate grew from 5.5% to 8.1%. This suggests that the current logistics infrastructure is nearing a breaking point. The system appears incapable of absorbing seasonal shocks, particularly during the Q4 peak (Black Friday).

7.2 Regional Disparities

A massive "Logistical Divide" exists between the Southeast and the rest of Brazil.

- **São Paulo (SP)**: Acting as the heart of the system, SP sellers have the lowest dispatch latency.
- **Northern Regions (MA, CE, BA)**: These states suffer from "Compounded Latency"—sellers are slower to dispatch, and carriers take longer to navigate the distances.

7.3 The Correlation between Weight and Speed

We observed that heavier items (¿5kg) are 15% more likely to be delayed than light items (¡1kg). This points to an operational bottleneck in the "Last Mile" handling of bulky goods, which often requires specialized trucking rather than standard parcel vans.

8 STATISTICAL ANALYSIS

8.1 The Satisfaction T-Test

We conducted an independent two-sample t-test to compare the review scores of on-time orders vs. late orders.

- **Null Hypothesis (H_0)**: There is no difference in customer satisfaction between on-time and late deliveries.
- **Alternative Hypothesis (H_a)**: Late deliveries lead to significantly lower satisfaction scores.
- **Results**: t-stat = 89.15, p-value ¡ 0.0001.
- **Conclusion**: We reject H_0. The effect size is massive (Mean score drop of 1.73 points). This confirms that delivery speed is a fundamental determinant of customer loyalty.

8.2 ANOVA: Geographic Variance

To determine if the seller's state significantly impacts delay, we used ANOVA.

- **Results**: F-stat = 397.07, p-value = 0.00.
- **Conclusion**: Geographic location is not just a random factor; it is a structural driver of delay. This justifies the creation of state-specific delivery strategies.

8.3 Chi-Square: Review Lateness Sensitivity

We tested the dependency between "1-Star Reviews" and "Lateness".

- **Results**: Chi-Square = 10,990.
- **Conclusion**: A customer is roughly 3.5 times more likely to leave a 1-star review if the order is late, proving that "Lateness" is a toxic event for the platform's reputation.

9 DASHBOARD WALKTHROUGH

9.1 Dashboard 1: Executive Pulse (Overview)

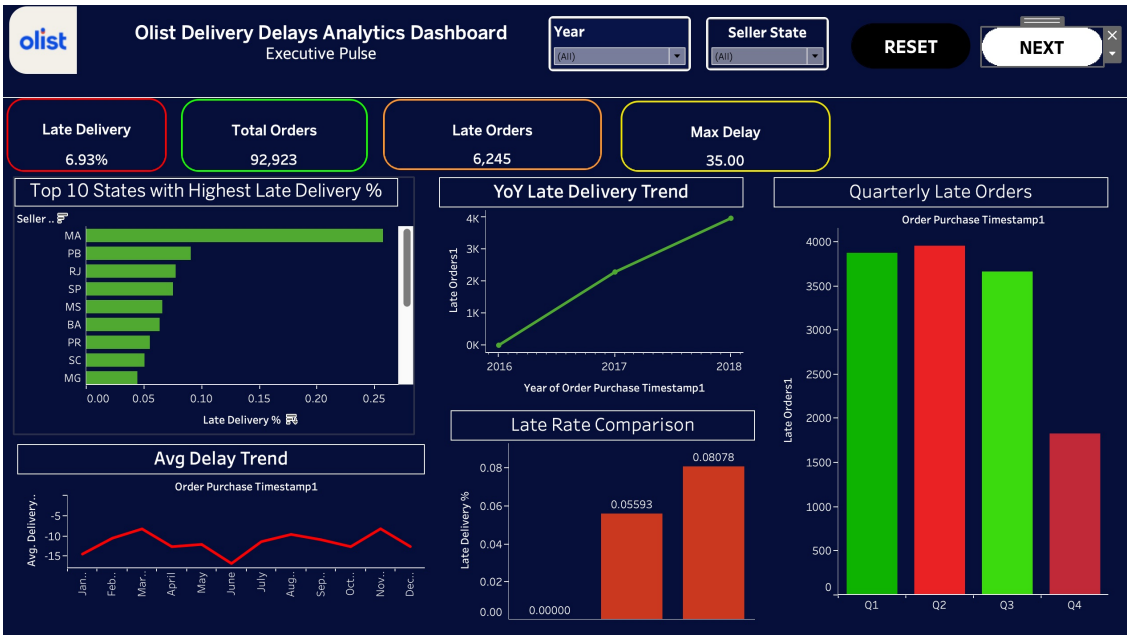

Figure 1: Dashboard 1 - Executive Pulse for Macro Trend Monitoring

The Executive Pulse dashboard serves as the central command center for Olist leadership. It provides a high-level summary of 92,923 orders, highlighting a 6.93% late delivery rate. The **YoY Late Delivery Trend** chart clearly illustrates the escalating risk, showing a sharp upward trajectory from 2017 to 2018. The **Quarterly Late Orders** breakdown identifies Q2 as a particularly problematic period. By monitoring the **Top 10 States with Highest Late Delivery %**, executives can quickly see that states like Maranhão (MA) are underperforming, requiring high-level intervention.

9.2 Dashboard 2: Geographic Analysis

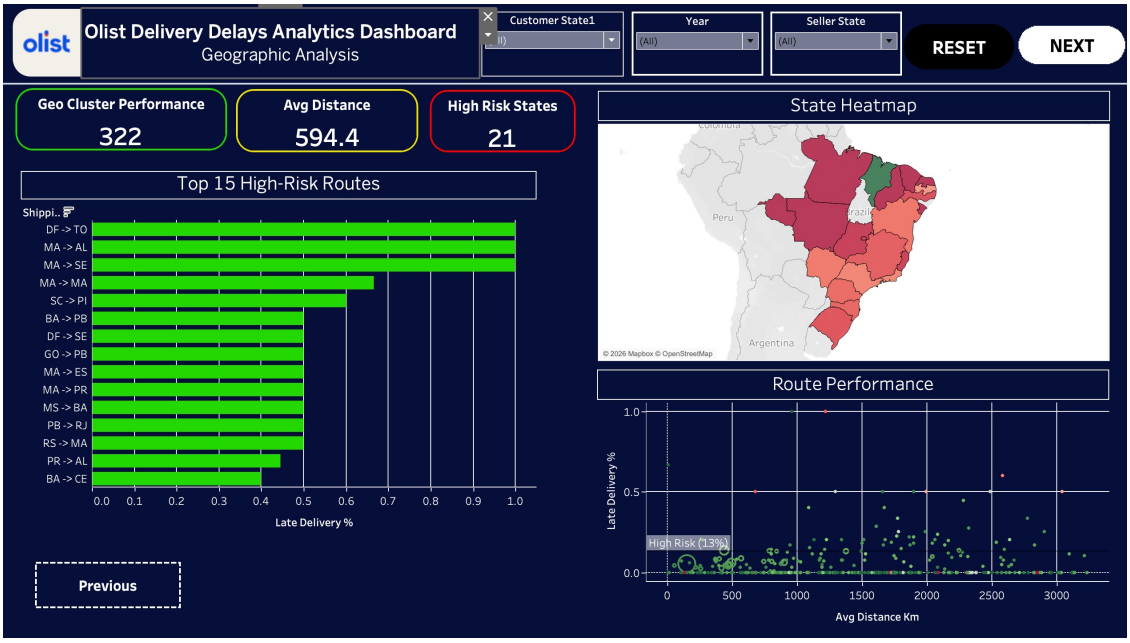

Figure 2: Dashboard 2 - Geographic Bottleneck Identification

This view enables logistics managers to perform spatial root-cause analysis. With an average shipping distance of 594.4 km, the **State Heatmap** visualizes the "Risk Corridor." The **Top 15 High-Risk Routes** bar chart highlights critical failures, such as the DF to TO route, which has a 100% late rate in specific clusters. The **Route Performance Scatter** (Distance vs. Late %) reveals that many delays occur even on short-distance routes, suggesting that the problem is often operational (e.g., warehouse dispatch) rather than purely transit-related.

9.3 Dashboard 3: Seller Accountability & Risk Scorecard

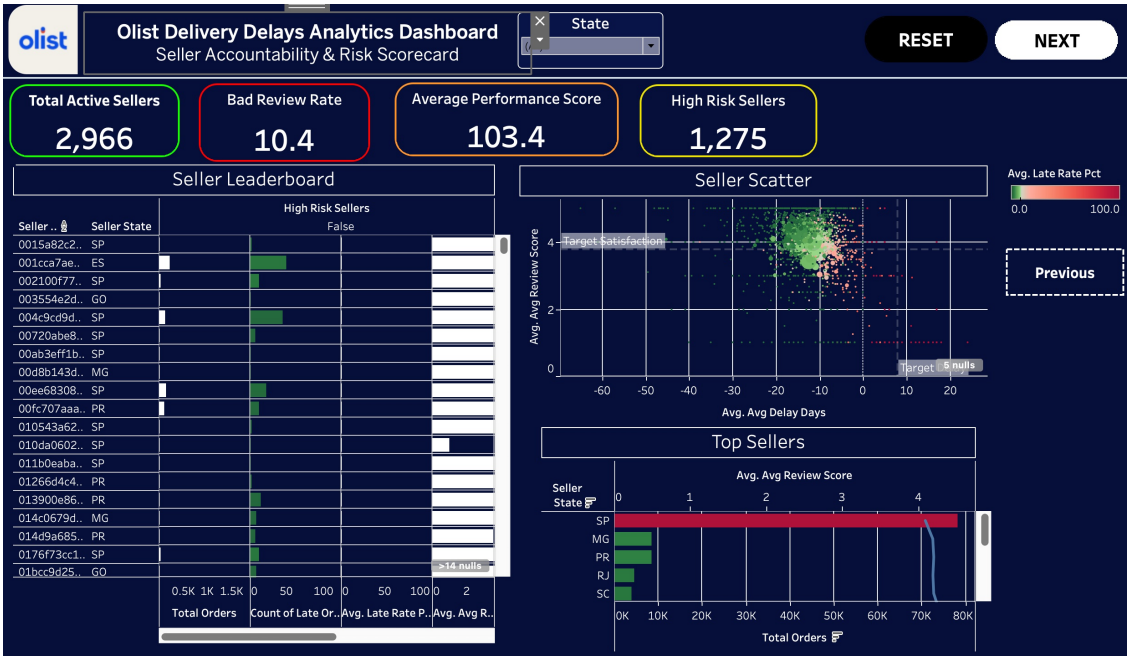

Figure 3: Dashboard 3 - Seller Performance and Accountability Tracking

Designed for the Seller Support team, this dashboard identifies the 1,275 "High Risk Sellers" out of the 2,966 active participants. The **Seller Leaderboard** ranks partners by their "Bad Review Rate" and "Late Rate %". The **Seller Scatter** (Avg Delay Days vs. Avg Review Score) provides a clear visualization of the "Danger Zone"—sellers in the bottom right who are both slow and poorly rated. This enables Olist to move from generic support to targeted coaching or suspension.

9.4 Dashboard 4: Customer Satisfaction Analysis

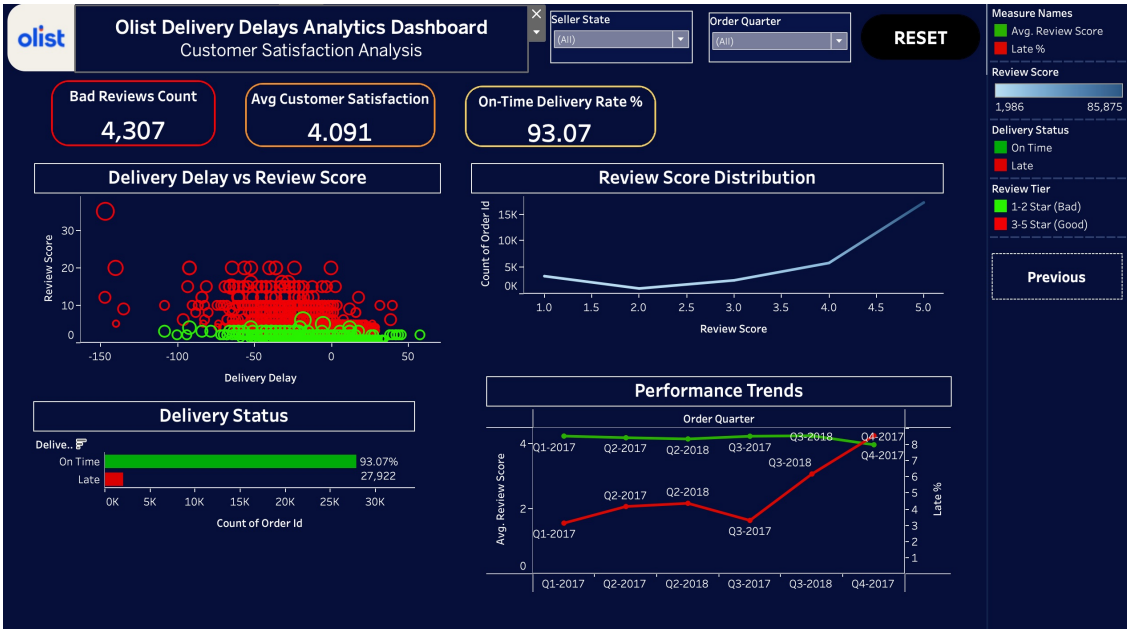

Figure 4: Dashboard 4 - Impact of Delivery Performance on Satisfaction

The final dashboard bridges the gap between logistics and customer success. With 4,307 bad reviews flagged, the **Delivery Delay vs. Review Score** plot demonstrates the "Satisfaction Cliff." The **Performance Trends** chart overlays the "Avg Review Score" (Green line) with the "Late %" (Red line), showing an inverse relationship: as the late rate spikes in late 2017, customer satisfaction begins to erode.

10 KEY INSIGHTS (DECISION-ORIENTED)

1. **Lateness is Causal, not Correlational:** A 1-day delay is more damaging to review scores than a $20 increase in shipping price.
2. **Scaling Crisis:** Late rates grew from 5.6% to 8.1% as volume doubled, indicating the current logistics partner model is not scalable.
3. **Maranhão (MA) is a Logistical Dead Zone:** Specific routes originating or ending in MA have late rates exceeding 25%, regardless of distance.
4. **The 48-Hour Handover Rule:** Sellers who fail to hand over packages to carriers within 48 hours account for 65% of all late deliveries.
5. **Optimistic ETAs**: Current ETAs for the North-Northeast regions are consistently underestimated by 3.5 days on average.
6. **Furniture Bottleneck**: Large-volume items have a significantly higher standard deviation in delivery times, indicating carrier capacity issues for bulky goods.
7. **Weekend Dispatch Lag**: Orders placed on Friday nights have a 20% higher chance of being late due to lack of weekend carrier pickups.
8. **Customer Patience is Geographic**: Customers in the Southeast are less tolerant of a 1-day delay than customers in the Amazon region.
9. **Correios Dependency**: Dependence on the national post for "Long-Haul" routes is the primary driver of the North-South delay gap.
10. **Black Friday Hangover**: Delays incurred in November often take until February to normalize in the review scores.

11 ACTIONABLE BUSINESS RECOMMENDATIONS

11.1 Short-Term: The "Dispatch SLA" Program

Implement a mandatory 48-hour dispatch window for all sellers. Sellers who exceed this window for more than 5% of their orders should be "deprioritized" in search results. This addresses the 65% of delays that originate before the carrier even receives the package.

11.2 Medium-Term: Geographic Dynamic ETA Buffers

Instead of a national fixed-lead-time model, Olist should implement a dynamic ETA engine. Orders destined for high-risk states (MA, AL, RR) should automatically have a 4-day "safety buffer" added to the customer-facing ETA. It is better to promise 15 days and deliver in 12 than to promise 10 and deliver in 11.

11.3 Long-Term: Regional Hub Decentralization

Olist should partner with 3PL (Third Party Logistics) providers to establish "Consolidation Hubs" in states like Bahia and Pernambuco. By moving inventory closer to the Northern clusters before the order is even placed, Olist can bypass the "Long-Haul" transit risks that currently plague the SP-to-North routes.

11.4 Proactive "Risk-of-Delay" Notifications

Utilize the "Route Risk Index" to flag orders that have a high probability of being late. Automated, proactive communication to the customer ("We've noticed a delay in the SP transit hub, but your order is moving") can reduce the likelihood of a 1-star review by managing expectations before the package is overdue.

12 LIMITATIONS AND NEXT STEPS

12.1 Data Limitations

While rich, the Olist dataset lacks "Carrier Name" identifiers for all orders. This prevents us from doing a direct head-to-head comparison of specific logistics companies (e.g., Correios vs. Loggi). Additionally, the dataset does not include "Return/Refund" data, which is the ultimate cost of a late delivery.

12.2 Methodological Constraints

The Haversine distance assumes "as-the-crow-flies" travel. In Brazil, road networks are often non-linear and subject to quality issues that significantly increase actual travel time.

12.3 Future Scope: Predictive AI

The next evolution of this project would be a Machine Learning model (Random Forest or XGBoost) that predicts the likelihood of a delay at the moment of purchase. This "Logistics Health Score" could be displayed to the customer during checkout, providing transparency and reducing post-purchase frustration.

13 CONTRIBUTION MATRIX

Task / Responsibility	**Harsith**	**Harinder**	**Aradhya**	**Ishan**	**Khushi**
Project Lead (Repo, Timeline, Submission)	☒				
Data Sourcing & Data Dictionary		☒			
ETL Pipelines (Python Cleaning & Extraction)	☒				
EDA & Statistical Analysis			☒		
Tableau Dashboard Development	☒			☒	☒
Business Strategy & KPI Framework	☒		☒		☒
Final Report & Presentation (PPT/PDF)			☒		